

BEGINNER

pedro



STEM Program

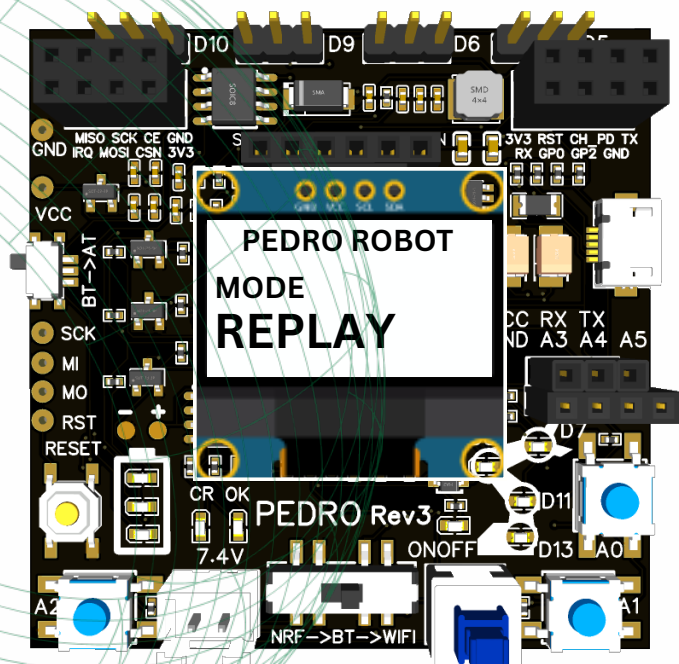
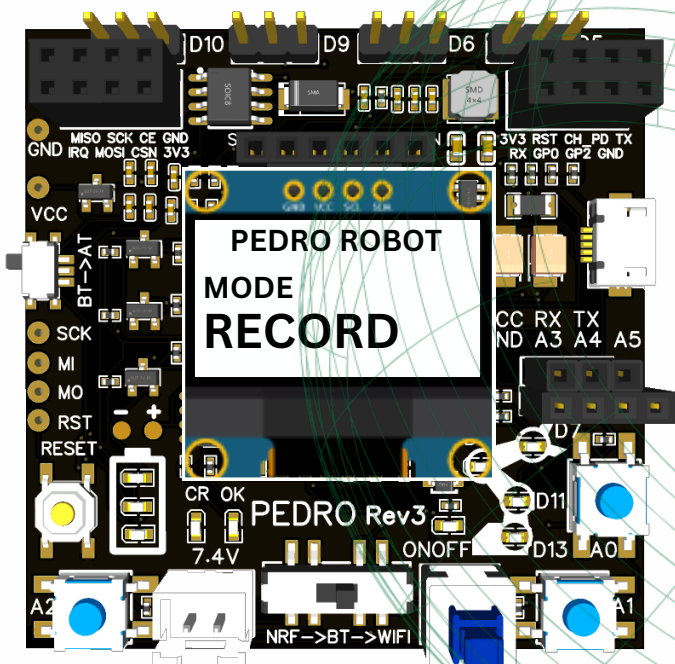
Lesson #7: Replay & Record Mode



Record



Replay



Complete this lesson
to earn your Pedro
Badge



Add it to your Pedro
Passport to track
your STEM journey

YOUR MISSION: Learn how robots record its movements and replay them automatically.

Learning Goals

In this lesson, students discover one of the most exciting features of robotics:

Instead of writing code line by line, students physically move the robot, record its movements, and replay them automatically. This lesson introduces important robotics concepts such as:

- memory
- automation
- motion sequences
- data recording
- repeatability



At the end of this lesson, you will unlock your Pedro Badge.

🏆 the Pedro Replay Badge

Teacher Notes

- No prior robotics experience required.
- Duration: 60–90 minutes
- Difficulty: ● Beginner
- Required Materials
 - **1 Assembled** Pedro robot **per student or per group** (*recommended: groups of 2 students*)
 - **OLED Screen 128X64** and **USB cable**
 - **1 computer** with the Arduino IDE installed



Think: The **brain icon** encourages students to make careful observations, share their ideas, and engage in discussion by explaining and justifying their answers.



Learn: The **light bulb icon** indicates important concepts that students should read carefully. Students are encouraged to ask questions and discuss these topics about the robotics concepts presented in the lesson.



Make: The **pencil icon** indicates that students must complete the activity on the page. These activities are designed to reinforce the concepts learned through experimentation, and problem-solving.



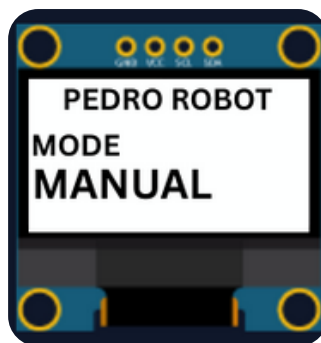
Board Configuration

(Skip this activity if the Pedro program is already uploaded)

The **Pedro Program** is the core software that powers and controls the robot. It acts as the robot's "brain," managing all behaviors and interactions. Through this single program, students can access and explore multiple control modes, allowing them to understand how a robot receives instructions, processes them, and translates them into physical actions.

1. **Insert** the Oled Screen into the Pedro board
2. **Connect** the board to your computer with the Micro USB cable.
3. Open **Arduino IDE**
4. **Install the required libraries** from the Library Manager:
 - Tools → Manage Libraries → search **PedroRobot** → Install
 - Tools → Manage Libraries → search **U8glib** → Install
 - Tools → Manage Libraries → search **RF24** → Install
5. Select the port that appear when you connect Pedro robot: **Tools → Port**
6. Select the board: **Tools → Board → Arduino AVR Boards → Arduino Micro**
7. Open the PedroRobot program: **File → Examples → PedroRobot → Pedro**
8. **Compile and upload** the program to your Pedro board.

✓ If everything is correct, Pedro's OLED screen will display **MANUAL MODE**.



Press **A0** and observe what happens. Each time you press **A0**, Pedro selects the next servo. The LED corresponding to the selected servo lights up, indicating which part of the robot you are controlling. The mapping between the LEDs and the servos is shown below:

- LED D13 → servo D5 (Base)
- LED D11 → servo D6 (Shoulder)
- LED D8 → servo D9 (Elbow)
- LED D7 → servo D10 (Gripper)



What Is Inverse Kinematics?

Imagine that you want a robot arm to pick up an object placed on a table. As a human, you simply think: "**Move your hand to the object.**"

You do not think about how much your shoulder, elbow, or wrist must rotate. For a robot, however, this is much more complicated. A robot cannot directly understand where its hand should go. Instead, it must calculate the exact angle of every joint required to reach the desired position.

This calculation is called **Inverse Kinematics (IK)**. There are two ways to describe the movement of a robot arm.

Forward Kinematics

The user specifies the angle of each joint. For example:

- Base = 30°
- Shoulder = 45°
- Elbow = 90°

The robot calculates where the end of the arm (the end effector) will be.

Joint Angles → Robot Position

Inverse Kinematics

The user specifies the desired position of the robot's hand. For example:

X = 120 mm

Y = 80 mm

The robot calculates the angle of every joint needed to reach that position.

Target Position → Joint Angles

Inverse kinematics allows robots to automatically reach a point without the user controlling each servo individually. Today, inverse kinematics is used in:

-  Industrial robot arms
-  Automated manufacturing
-  Surgical robots
-  Space robots
-  Warehouse robots



Understanding Record & Replay Mode

Pedro doesn't use **inverse kinematics** because it's powered by 360° continuous rotation servos (MG90S) not classic positional servos. That means Pedro doesn't « go to » a position it just spins forward or backward, depending on the PWM signal.

To perform a Repeat sequence instead of recording positions, Pedro records three specifics data of each selected servo:

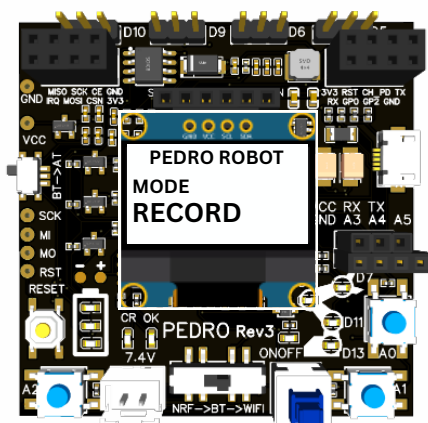
- ⌚ **the time**
- 🌀 **the direction**
- ⚡ **the speed**

The result? Pedro can **replay movements** like a macro recorder just by following the same timing and motion patterns. Each movement becomes part of a motion sequence.

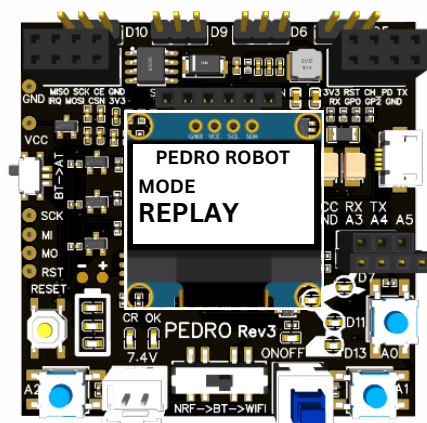
Replay Mode allows Pedro to reproduce the recorded sequence automatically. The robot repeats the exact same movements without human control. This creates an automated robotic behavior. Students discover that:

💡 Robots can repeat tasks consistently. This is one of the main advantages of robotics in industry.

Record Movement

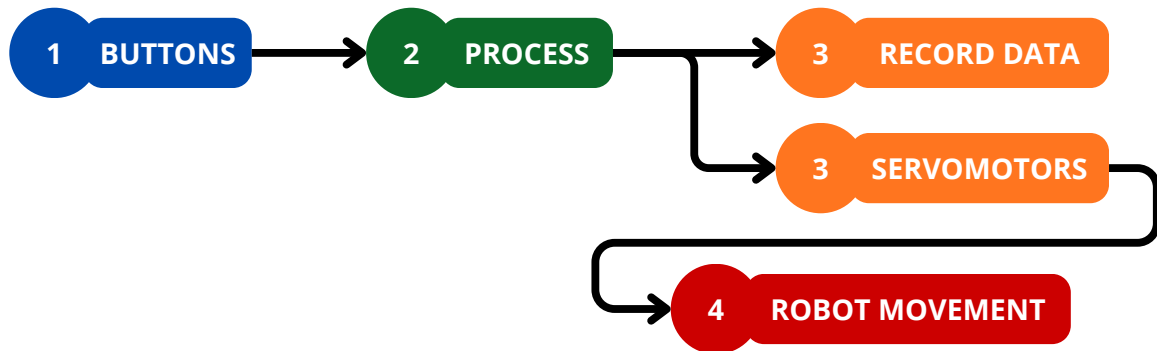


Replaying Movement

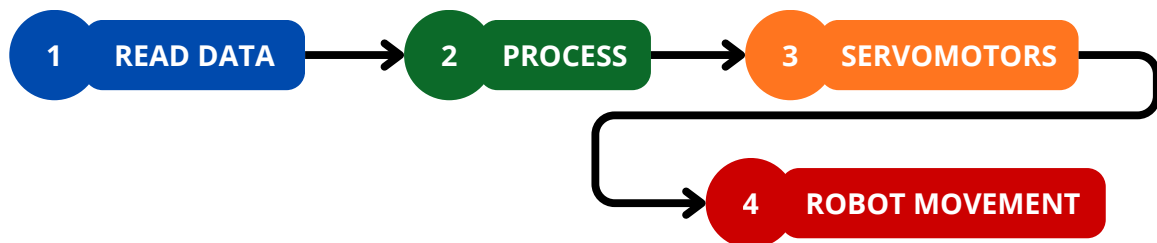


Why is repeatability important in robotics? Why do factories use automated robots? What happens if incorrect data is recorded? How does Pedro “remember” movements? Can robots learn by observing humans?

The Recording chain is:



The Replay chain is:



SRAM – Temporary Working Memory (2.5 KB)

As seen in the Lesson #4 Pedro robot uses an **ATmega32U4 microcontroller** which contains several types of memory, each designed for a specific purpose. The ATmega32U4 has the following memory resources:

Memory Type	Size	Main Function
Flash Memory	32 KB	Stores the compiled robot program (firmware)
SRAM	2.5 KB	Stores temporary data while the robot is running
EEPROM	1 KB	Stores permanent user data and recorded sequences

The **2.5 KB SRAM** is used while Pedro is running and It stores temporary

information. In Pedro's **Repeat Mode**, **SRAM** stores the recorded movement sequences.

As seen earlier Pedro does not store servo positions because the MG90S servos are **360° continuous rotation servos**. Instead, each movement is stored using three parameters:

Data Stored	Example
 Time	2 seconds
 Direction	Forward / Reverse
 Speed	PWM value

Example of stored movement:

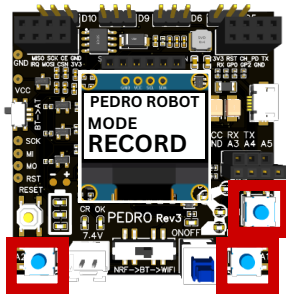
Servo: Base, **Direction:** Forward, **Speed:** 60, **Time:** 2000 ms

💡 During Replay Mode, Pedro reads these values from SRAM and reproduces the same sequence automatically.



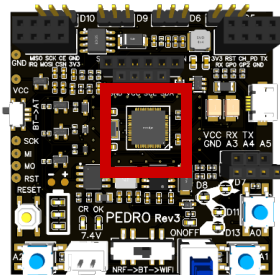
How Pedro Records a Motion

1 BUTTONS



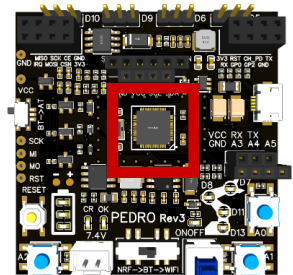
Control the Pedro Transmitter using buttons **A0**, **A1** and **A2**

2 PROCESS



The **microcontroller** process the data and generates the appropriate **PWM signal**.

3 RECORDING



The program **records** the servo data including: **Time**, **Direction** and **Speed**

3 SERVO MOTORS



PWM Signal



The selected servo receives the **PWM signal** and rotates in the requested direction.

4 ROBOT MOVEMENT



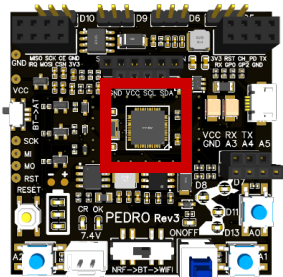
The selected servo rotates, causing the robot to perform the requested movement.



How Pedro Replay a Motion

1

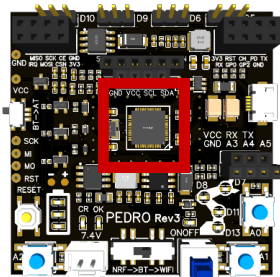
READ STORED DATA



The **microcontroller** read the data stored of each servo: **time**, **direction** and **speed**

2

PROCESS

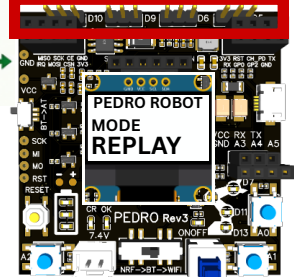


The **microcontroller** process the data and generates the appropriate **PWM signal**.



3

SERVO MOTORS



The selected servo receives the **PWM signal** and rotates in the requested direction.

4

ROBOT MOVEMENT

The selected servo rotates, causing the robot to perform the requested movement.





Recording & Replaying Robot Movements

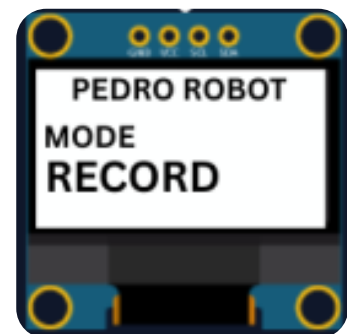
Pedro uses a simple logic system in three steps:

1. **Entering Record Mode**
2. **Recording a Sequence**
3. **Entering Replay Mode**

STEP 1: Entering Record Mode

1. Power ON the robot
2. Enter the **Select Mode** menu (hold **A0** for 4s)
3. Select **RECORD mode** from the menu
4. Press **A0** to confirm
5. The OLED screen displays:

💡 Note: Pedro is now ready to memorize movements.



STEP 2: Recording a Sequence

You will now create your first robotic routine. **The direction of rotation is not important.** Simply complete the following sequence:

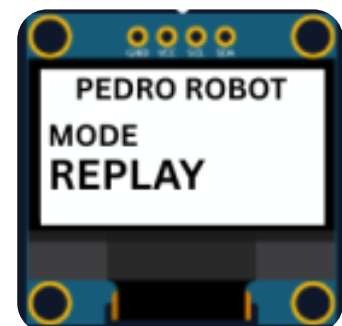
1. Rotate base
2. Lift arm
3. Open gripper
4. Lower arm
5. Close gripper

💡 Note: Each movement has been stored inside Pedro's **SRAM memory**.

STEP 3: Entering Replay Mode

1. Enter to the **Select Mode** menu (hold **A0** for 4s)
2. Select **REPLAY mode** from the menu
3. Press **A0** to confirm
4. The OLED screen displays:

💡 Note: Pedro now repeats the recorded actions continuously.

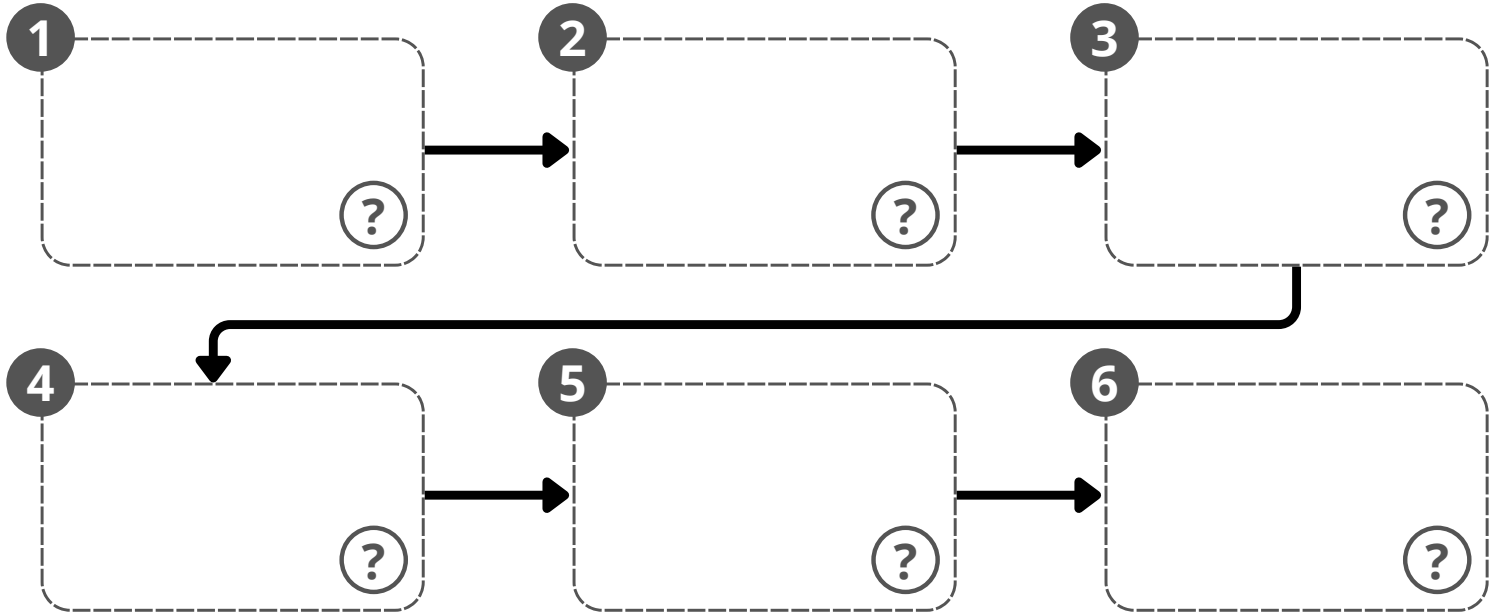


To stop the replaying sequence press A0 until the Select mode menu display.



Replay Chain

Write the correct letters in boxes 1 to 6 to complete the Record chain.



A Servo Moves

The servo interprets this PWM signal and moves to the requested position.

B Microcontroller reads button

The microcontroller detects each button pressed

C ATmega32u4 store data in memory

The microcontroller store servo data: **time, direction, speed**

D Microcontroller generates a PWM signal

Send the PWM signal to chosen servo

E Loop the sequence

Repeat the recording chain for the next sequence

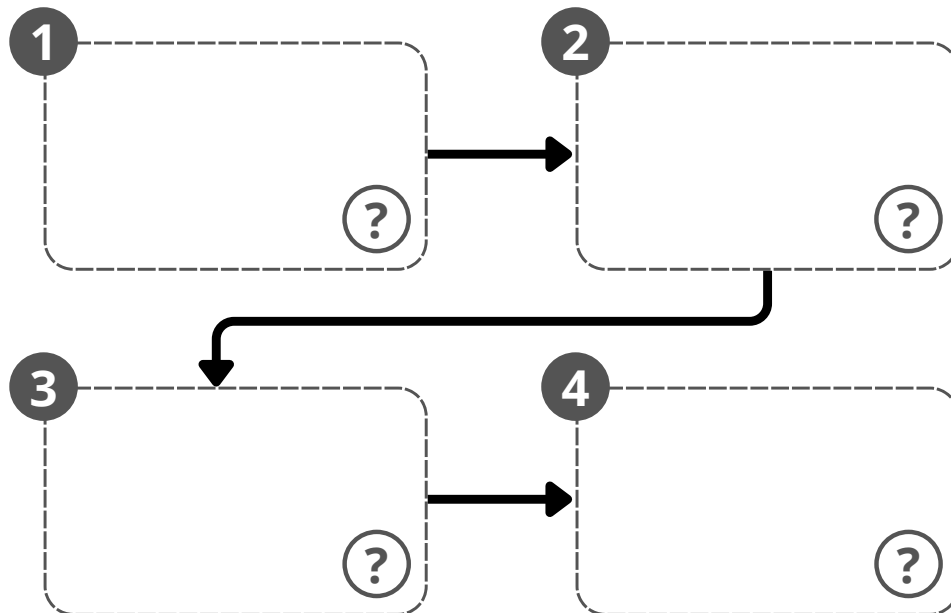
F You click a button (Select Base + A1)

Choose which servo and movement



Replay Chain

Write the correct letters in boxes 1 to 4 to complete the Replay chain.



A Servo Moves

Servo receives the PWM signal and moves to the requested position.

B Microcontroller decode data

The microcontroller decodes the data and send signal to the servo

D Loop Replay Sequence

Loop the chain for the next movement

E ATmega32u4 reads data in memory

Read the data stored for each sequence: **time, direction, speed**



Quiz - Check Your Understanding

1. What is the purpose of Record Mode?

- ☐ Charge the battery
- ☐ Store robot movements
- ☐ Connect to WiFi
- ☐ Control LEDs

2. What does Replay Mode do?

- ☐ Deletes movements
- ☐ Replays stored actions automatically
- ☐ Turns OFF the robot
- ☐ Resets the program

3. Which type of memory does Pedro use for recording mode?

- ☐ EEPROM
- ☐ Flash Memory
- ☐ SRAM
- ☐ Hard disk

4. Which data are recorded during Record Mode?

- ☐ Servo positions only
- ☐ Time, direction, and speed
- ☐ Battery voltage
- ☐ OLED brightness

5. Why doesn't Pedro use inverse kinematics?

- ☐ The battery is too small
- ☐ Pedro uses continuous rotation servos
- ☐ The OLED screen is too small
- ☐ Bluetooth is disabled

6. What happens when Pedro enters Replay Mode?

- ☐ It records new movements
- ☐ It repeats stored movements automatically
- ☐ It turns OFF the servos
- ☐ It resets the memory

7. What is one major advantage of robotics in industry?

- ☐ Robots can sleep
- ☐ Robots can repeat tasks consistently
- ☐ Robots can print documents
- ☐ Robots can create batteries

8. Which button is used to access the Select Mode menu?

- ☐ A1
- ☐ A2
- ☐ A0
- ☐ RESET

9. What engineering concept is tested when replaying movements multiple times?

- ☐ Battery charging
- ☐ Repeatability
- ☐ Bluetooth pairing
- ☐ Screen resolution

10. What is the main purpose of robotic automation?

- ☐ Make robots colorful
- ☐ Allow robots to perform tasks automatically
- ☐ Increase screen brightness
- ☐ Charge batteries faster

Congratulations Mission Completed!

You have unlocked: 🏆 the Pedro Replay Mode badge
Don't forget to stick your Pedro Badge into your Pedro Passport before starting the next lesson.

